STALKS OF THE GLUEING KERNEL AND COKERNEL UNDER INTEGRAL UNIVERSALLY-INJECTIVE FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1881-1896 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 1898-1928, the complete author proof (29 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 33 of queue_spaces-pushouts.json (pb=813, stmt 1881-1896, proof 1898-1928).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0AEQ: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 1840–1850. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : Y \to X$ be a morphism of algebraic spaces over S. Let $Z \subset X$ closed subspace such that $f^{-1}Z \to Z$ is integral and universally injective. Let $\overline{y}$ be a geometric point of Y and $\overline{x} = f(\overline{y})$. Let $\mathcal{G}$ be an abelian sheaf on Y. Then the map of two term complexes*

$$(f_*\mathcal{G}_{\overline{x}} \to (f \circ j')_*(\mathcal{G}|_V)_{\overline{x}}) \longrightarrow (\mathcal{G}_{\overline{y}} \to j'_*(\mathcal{G}|_V)_{\overline{y}})$$

induces an isomorphism on kernels and an injection on cokernels. Here $V = Y \setminus f^{-1}Z$ and $j' : V \to Y$ is the inclusion.

Proof. Choose a distinguished triangle

$$\mathcal{G} \to Rj'_*\mathcal{G}|_V \to Q \to \mathcal{G}[1]$$

n $D(Y_{étale})$. The cohomology sheaves of Q are supported on $|f^{-1}Z|$. We apply Rf_* and we obtain

$$Rf_*\mathcal{G} \to Rf_*Rj'_*\mathcal{G}|_V \to Rf_*Q \to Rf_*\mathcal{G}[1]$$

Taking stalks at $\overline{x}$ we obtain an exact sequence

$$0 \to (R^{-1}f_*Q)_{\overline{x}} \to f_*\mathcal{G}_{\overline{x}} \to (f \circ j')_*(\mathcal{G}|_V)_{\overline{x}} \to (R^0f_*Q)_{\overline{x}}$$

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

We can compare this with the exact sequence

$$0 \to H^{-1}(Q)_{\overline{y}} \to \mathcal{G}_{\overline{y}} \to j'_*(\mathcal{G}|_V)_{\overline{y}} \to H^0(Q)_{\overline{y}}$$

Thus we see that the lemma follows because $Q_{\overline{y}} = Rf_*Q_{\overline{x}}$ by Lemma 0AEQ. $\square$